Binary Search

Definition

Binary Search is a searching algorithm used to find an element in a **sorted array**. It repeatedly divides the search range into two halves.

Algorithm

1. Start with `low = 0` and `high = n - 1`.
2. Find the middle element:
   `mid = (low + high) // 2`
3. Compare the middle element with the element to be searched.
4. If `arr[mid] == key`, the element is found.
5. If `key > arr[mid]`, search the **right half**.
6. If `key < arr[mid]`, search the **left half**.
7. Repeat until the element is found or `low > high`.
8. If `low > high`, the element is not present.

Time Complexity

Case	Complexity
Best Case	**O(1)**
Average Case	**O(log n)**
Worst Case	**O(log n)**

Space Complexity

- **Iterative Binary Search: O(1)**
- **Recursive Binary Search: O(log n)** due to the recursion stack.

Important: Binary search works correctly only when the input array is **sorted**.